



Generative and Language Models

Luiz Eugênio Santos Araújo Filho

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Summary

- 1 Introduction
- 2 VAE
- 3 GAN
- 4 Diffusion models
- 5 Attention in language models
- 6 Transformers

Which one is fake?



A

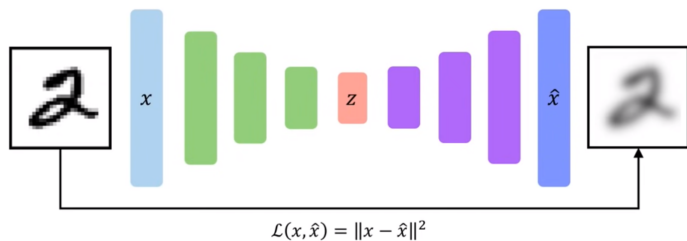


B

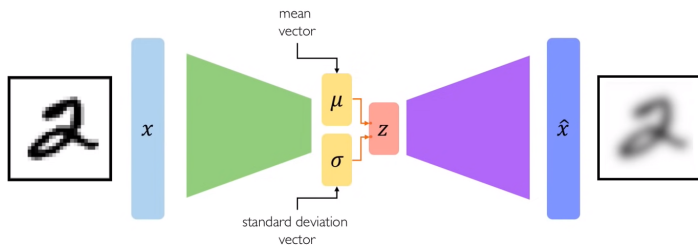


C

Autoencoders

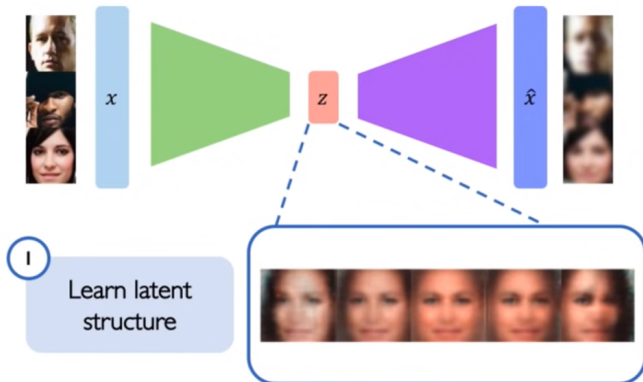


Variational Autoencoders

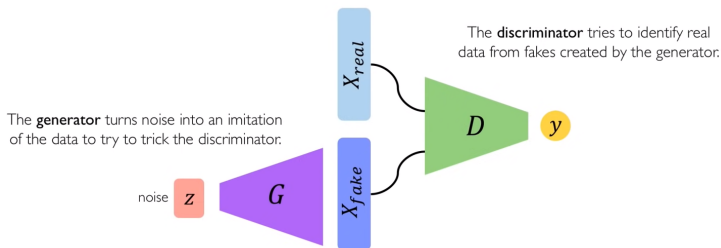


Variational Autoencoders

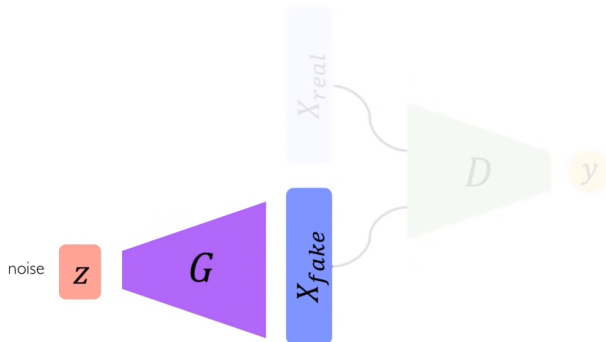
Learn latent space



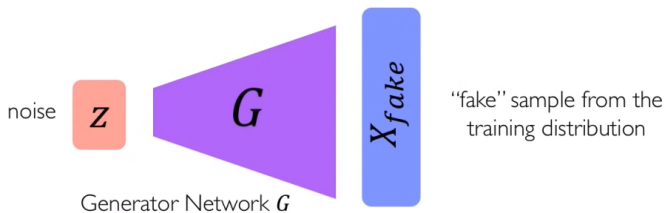
Generative Adversarial Networks



Generative Adversarial Networks



Generative Adversarial Networks



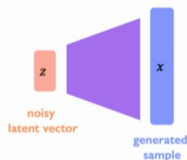
Generative Adversarial Networks

Example: CycleGAN

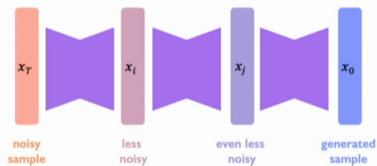


Diffusion models

VAEs/GANs: Generating samples in one-shot directly from low-dimensional latent variables

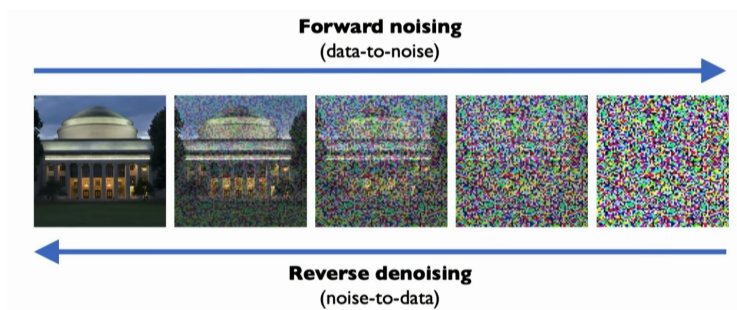


Diffusion: Generating samples iteratively by repeatedly refining and removing noise



Diffusion models

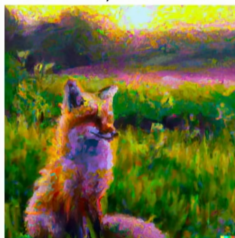
Diffusion process



Diffusion models

Text-to-image generation

"a painting of a fox sitting in a field at sunrise in the style of Claude Monet"



"an ibis in the wild, painted in the style of John Audubon"

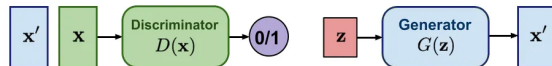


"close-up of a snow leopard in the snow hunting, rack focus, nature photography"

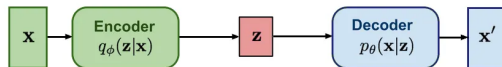


Generative models summary

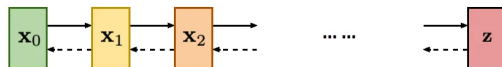
GAN: Adversarial training



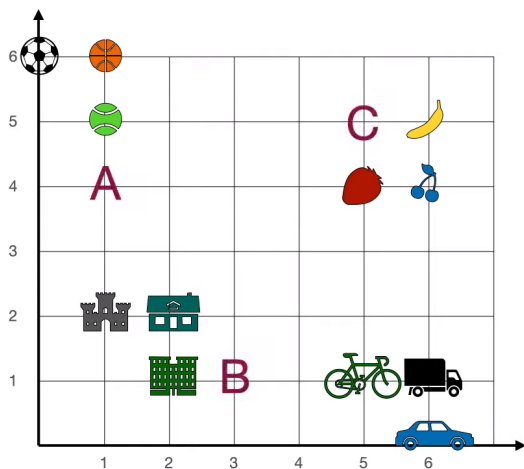
VAE: maximize variational lower bound



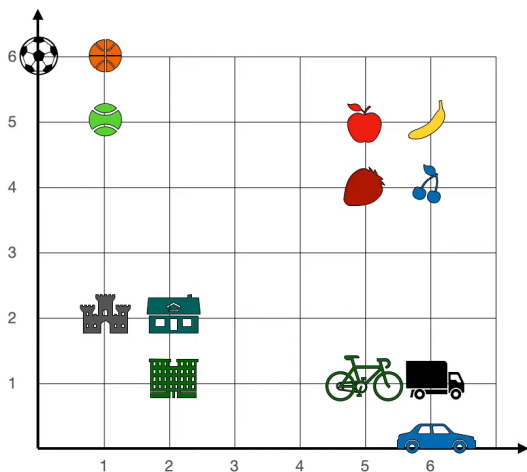
Diffusion models:
Gradually add Gaussian noise and then reverse



Attention mechanism



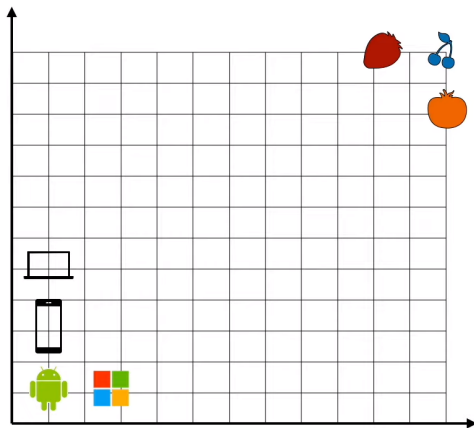
Attention mechanism



Attention mechanism

Please buy an **apple** and an orange.

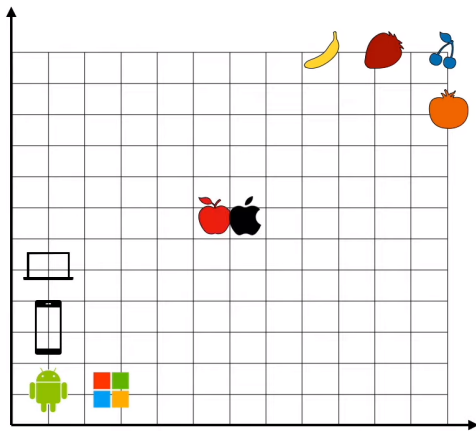
Apple unveiled the new phone.



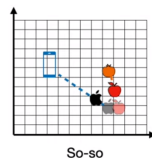
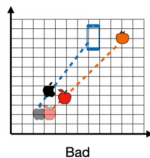
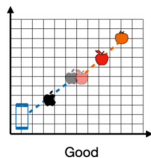
Attention mechanism

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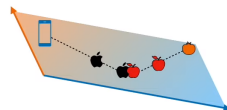
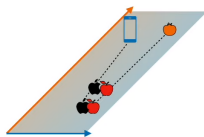
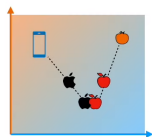
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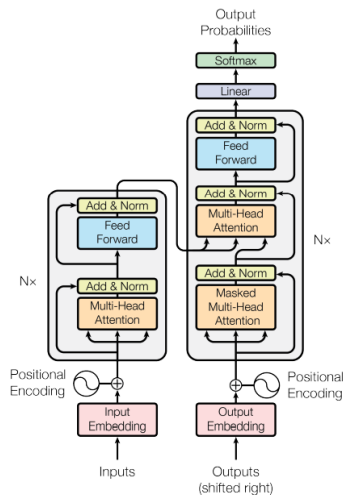
Attention mechanism



Attention mechanism



Transformer architecture



Transformer architecture

